

LECTURE 9: INTRODUCTION TO OLIGOPOLIES

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WEEK 4

WE CAN START APPLYING OUR GAME THEORY TO FIRM COMPETITION

- With monopolies, we studied when one firm has significant barriers of entry and has full market power
- Oligopolies have this same principle, but now there are multiple firms
 - Our firms are going to compete with each other and act at the same time, so we need to use our ideas of game theory
 - Our firms are also going to try to maximise their profits, so we need our ideas from monopolies



WHAT WE'RE GOING TO STUDY ARE "MODELS" OF THE WORLD

- We're going to use some simple mathematical models of how firms compete
- These aren't designed to capture every nuance in real-world competition
- For certain industries and markets they can surprisingly be predictive of actual outcomes
- We're going to see how outcomes end up somewhere between monopolies (high prices, low quantities) and perfect competition (prices at MC)

WHAT WE'RE GOING TO STUDY ARE "MODELS" OF THE WORLD

- Three key models for us to study - all variations on the same theme
- First two are like our simultaneous games - firms can compete on price or quantity
- Third model is like our first mover/sequential games - one firm enters the market first

WHY DO WE USE MODELS?

- Real life competition is difficult, complex and dynamic
- Imagine you're an economist working at an airline, trying to work out whether selling more seats from Sacramento to LA will increase profits or not?
- What are some questions we might want to ask and answer?
- How will increasing the number of flights change prices? How will competitors respond? Will this increase prices or not?
- Models give us a framework that we can build on to use statistics, make predictions, have an intuitive understanding of the market to start to answer these questions

WHEN IS A MODEL A GOOD MODEL?

- We started this course studying the bookends of competition: perfect competition and monopolies
- Our models should give the monopoly outcome when we model 1 firm
- Our models should get closer to the perfectly competitive outcome when we model many firms!
- Our models should give us predictions that: we can use to make business decisions; and verify and refine with statistical tools
- **No model is perfect** - they are only useful if they are useful

FIRST MODEL - FIRMS CHOOSE QUANTITY

- Our first model of oligopoly competition is when firms choose quantity simultaneously. (Think: **I already have a factory, how much should I produce?**)
- This is also known as the Cournot model
- Increasing output of either firm decreases the price & margin for both firms - we assume they are both price takers
- We want to ask: **What is the Nash Equilibrium of output each firm should produce?**

FIRST MODEL - FIRMS CHOOSE QUANTITY

- Firms know the total market willingness to pay for flights $P = a - b \times Q_D$
- Each firm can compete to supply total demand: $Q_D = q_1 + q_2$
- The more each firm tries to sell, the more it will lower the price! (**A trade-off!**)
- Each firm will have a **best response** function on how much to sell given how much the other firm will sell!

COURNOT EXAMPLE - AIRLINES

This is how we approach solving Cournot games using both our profit-maximising math and our game theory:

- 1 - Find market demand as a function of how much each firm supplies
- 2 - Setup our profit maximisation problem for each firm - needs to consider the actions of the other firms!
- 3 - Find the best response functions - how much should each firm sell as a function of other firm's quantities
- 4 - Find where the best response functions coincide/intersect - this is a Nash-Cournot equilibrium

Let's do this on the board

COURNOT EXAMPLE - AIRLINES

- American Airlines and United are competing to sell flights from Sacramento to LA
- Inverse demand for this route is $P = 339 - Q_D$
- Both airlines have no fixed costs and constant marginal costs of \$147
- How many seats will each airline sell?



COURNOT EXAMPLE - AIRLINES

Step 1

The first thing to do, is when we see total market demand, Q_D replace this with the sum of quantities supplied by each firm: $q_U + q_A$

So the inverse demand curve is: $P = 339 - (q_U + q_A)$

COURNOT EXAMPLE - AIRLINES

We'll do step 2 and 3 for American Airlines first:

Step 2 Maximise profits

Each firm is going to maximise its profits with respect to how much it sells - we want to find when the derivative of profit is zero:

$$\begin{aligned}\max_{q_A} \pi_A(q_A, q_U) &= P \times q_A - 147 \times q_A \\ &= (339 - (q_U + q_A))q_A - 147 \times q_A \\ &= 339q_A - q_U \times q_A - q_A^2 - 147 \times q_A\end{aligned}$$

With first order conditions:

$$\begin{aligned}\frac{d\pi_A}{dq_A} &= 339 - q_U - 2q_A - 147 = 0 \\ 192 - q_U - 2q_A &= 0\end{aligned}$$

COURNOT EXAMPLE - AIRLINES

Step 3: Best response function

How much should American Airlines sell as a function of how much United sells? Or, what is American's profit maximising quantity given United is selling q_U flights

We start with: $192 - q_U - 2q_A = 0$

$$192 - q_U - 2q_A = 0$$

$$-2q_A = -192 + q_U$$

$$q_A = BR_A = 96 - \frac{1}{2}q_U$$

This is American's **best response** function to how many seats q_U sells - we know it's the optimal quantity conditional on q_U because we derived it from the profit maximising condition

COURNOT EXAMPLE - AIRLINES

We'll do step 2 and 3 for United:

Step 2 Maximise profits

Each firm is going to maximise its profits with respect to how much it sells - we want to find when the derivative of profit is zero:

$$\begin{aligned}\max_{q_U} \pi_U(q_A, q_U) &= P \times q_U - 147 \times q_U \\ &= (339 - (q_U + q_A))q_U - 147 \times q_U \\ &= 339q_U - q_U \times q_A - q_U^2 - 147 \times q_U\end{aligned}$$

With first order conditions:

$$\begin{aligned}\frac{d\pi_U}{dq_U} &= 339 - 2q_U - q_A - 147 = 0 \\ 192 - 2q_U - q_A &= 0\end{aligned}$$

COURNOT EXAMPLE - AIRLINES

Step 3: Best response function

How much should United sell as a function of how much American sells? Or, what is United's profit maximising quantity given American is selling q_A flights

We start with: $192 - 2q_U - q_A = 0$

$$192 - 2q_U - q_A = 0$$

$$-2q_U = -192 + q_A$$

$$q_U = 96 - \frac{1}{2}q_A$$

$$BR_U = 96 - \frac{1}{2}q_A$$

This is United's **best response** function - we know it's the optimal quantity given q_A because we derived it from the profit maximising condition

COURNOT EXAMPLE - AIRLINES

Finally, **step 4** - when each firms' best response functions intersect, the quantity is a Nash equilibrium.

It's both profit maximising conditional on other firm's behaviour, and no one wants to deviate.

COURNOT EXAMPLE - AIRLINES

How to do this? Simply substitute one best response function into the other:

We have:

$$(1) \quad q_A = 96 - \frac{1}{2}q_U$$

$$(2) \quad q_U = 96 - \frac{1}{2}q_A$$

Let's substitute (1) into (2)

$$q_U = 96 - \frac{1}{2} \left(96 - \frac{1}{2}q_U \right)$$

$$q_U = 96 - 48 + \frac{1}{4}q_U$$

$$\frac{3}{4}q_U = 48$$

$$q_U = 64$$

And we know that $q_A = 96 - \frac{1}{2}q_U$

So we get that $q_A = 64$

And prices will be: $P = 339 - (q_U + q_A) = 339 - 128 = \211

COURNOT EXAMPLE - AIRLINES

We know this is a Nash Equilibrium, because neither firm would want to deviate and change its output:

✅ American sells 64 → United's best response is to sell $96 - 0.5 \times 64 = 64$ → American's best response is to sell $96 - 0.5 \times 64 = 64$

👎 **Counterexample:** What if American sells 60 → United's best response is to sell $96 - 0.5 \times 60 = 66$ → American's best response is to sell $96 - 0.5 \times 66 = 63$ so **American deviates**

COURNOT EXAMPLE - AIRLINES

Compared to a monopoly - Cournot competition will lead to lower prices and higher output:

- **Cournot:** Price = \$211, Total quantity $Q_D = q_U + q_A = 128$
- **Monopoly:** Price = \$243, Total quantity $Q_D = 96$

In general, the more firms you keep adding, the closer you get to the outcome of a competitive market!

We'll keep it to two firms in this class to make the math easier

COURNOT, MONOPOLY AND PERFECT COMPETITIVE OUTCOME

Remember our mark up rule from monopolies:

$$P_{\text{Monopoly}} = MC \frac{1}{\left(1 + \frac{1}{\varepsilon}\right)}$$

There's the equivalent for Cournot competition - with n firms:

$$P_{\text{Cournot}} = MC \frac{1}{\left(1 + \frac{1}{n \times \varepsilon}\right)}$$

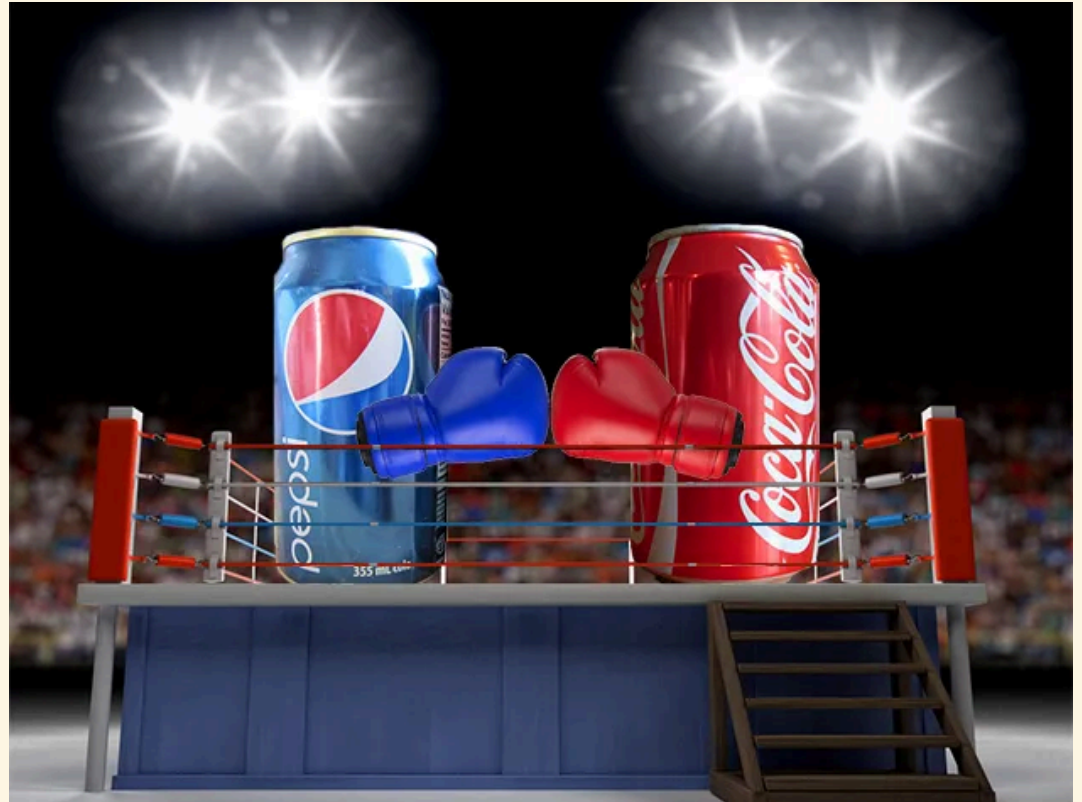
How many firms do we need to get to the perfectly competitive outcome?

$$P_{\text{Competitive}} = MC$$

Infinitely many - this is why we often say that perfect competition requires unrealistic assumption

THE POWER OF COURNOT IN REAL WORLD MARKETS

- There's one small addition we can add to our Cournot model to make it reality a little bit better
- **Production differentiation:**
Products made by competing firms might be substitutes with one another - but will be valued differently in the market
- Examples: iPhone versus Google Pixel; Coke v Pepsi; Nvidia v Intel GPUs
- Main change: Introducing separate demand curves for each competitor showing the willingness to pay for each differentiated product



DIFFERENTIATED COURNOT (CHOOSING QUANTITY WITH DIFFERING PRODUCTS)

In a week 1 lecture, I talked about how Nvidia is a monopolist for GPUs. This isn't quite true, there is a small amount of market share taken by Intel. Let's model this a bit more accurately.

Intel GPUs:

- Inverse demand: $P_I = 490 - q_N - 1.5q_I$
- Total costs: $TC_I = 40q_I$

Nvidia GPUs

- Inverse demand: $P_N = 150 - 0.3q_N - 0.1q_I$
- Total costs: $TC_N = 50q_N$

The key part is that the demand for each product depends on the other product, this links them as substitutes - ie, they're competing in related markets

If they act in Cournot competition (choosing quantity), what will their prices be?

DIFFERENTIATED COURNOT (CHOOSING QUANTITY WITH DIFFERING PRODUCTS)

The steps and math remain mostly the same, except that now each firm has its own demand curve

- 1 - Setup our profit maximisation problem for each firm - needs to consider the actions of the other firms! Each firm has their own inverse demand curve
- 2 - Find the best response functions - how much should each firm sell as a function of other firm's quantities
- 3 - Find where the best response functions coincide/intersect - this is a Nash-Cournot equilibrium

DIFFERENTIATED COURNOT (CHOOSING QUANTITY WITH DIFFERING PRODUCTS)

Let's start with finding Intel's profit and best response functions:

$$\begin{aligned}\max_{q_I} \pi_I &= \max_{q_I} P_I \times q_I - 40q_I \\ &= \max_{q_I} (490 - q_N - 1.5q_I) \times q_I - 40q_I \\ &= \max_{q_I} 490q_I - q_Nq_I - 1.5q_I^2 - 40q_I \\ &= \max_{q_I} 450q_I - q_Nq_I - 1.5q_I^2\end{aligned}$$

We optimise Intel's profit - setting the derivative to zero:

$$\begin{aligned}\frac{d\pi_I}{dq_I} &= 450 - q_N - 3q_I = 0 \\ 3q_I &= 450 - q_N \\ q_I &= BR_I = 150 - \frac{1}{3}q_N\end{aligned}$$

DIFFERENTIATED COURNOT (CHOOSING QUANTITY WITH DIFFERING PRODUCTS)

Now let's find Nvidia's profit and best response functions:

$$\begin{aligned}\max_{q_N} \pi_N &= \max_{q_N} P_N \times q_N - 50q_N \\ &= \max_{q_N} (150 - 0.3q_N - 0.1q_I) \times q_N - 50q_N \\ &= \max_{q_N} 150q_N - 0.3q_N^2 - 0.1q_I q_N - 50q_N \\ &= \max_{q_N} 100q_N - 0.3q_N^2 - 0.1q_I q_N\end{aligned}$$

We optimise Nvidia's profit - setting the derivative to zero:

$$\begin{aligned}\frac{d\pi_N}{dq_N} &= 100 - 0.6q_N - 0.1q_I = 0 \\ 0.6q_N &= 100 - 0.1q_I \\ q_N &= BR_N = 166.67 - 0.167q_I\end{aligned}$$

DIFFERENTIATED COURNOT (CHOOSING QUANTITY WITH DIFFERING PRODUCTS)

We have both of our best responses:

$$q_I = BR_I = 150 - \frac{1}{3}q_N$$
$$q_N = BR_N = 166.67 - 0.167q_I$$

Where they intersect is the Nash equilibrium: \begin{equation*}

$$q_I = 150 - \frac{1}{3}(166.67 - 0.167q_I)$$
$$q_I = 150 - 55.56 + 0.0556q_I$$
$$q_I = 94.44 + 0.0556q_I$$
$$0.9444q_I = 94.44$$
$$q_I = 100$$

\end{equation*}

So then, $q_N = 166.67 - 0.167 \times 100 = 150$

So even though both companies both make GPUs, differentiation in ability means Nvidia sells far more!

